SAVITRIBAI PHULE PUNE UNIVERSITY

A Mini Project Report based on:

SaaS over LAN

Submitted by:

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Under the Support and Guidance of

Prof. A. P. Bhalke

In partial fulfilment of

Cloud Computing Laboratory (310257)

COMPUTER ENGINEERING DEGREE OF

SAVITRIBAI PHULE PUNE UNIVERSITY 2024-2025

CERTIFICATE

This is to certify that following students from **Third Year Computer Engineering** have successfully completed and submitted the **Mini Project** work titled **SaaS Over LAN** at PDEA’s College of Engineering, Manjari (BK), Pune – 412307 in the partial fulfillment of the Bachelor’s Degree in Computer Engineering by Savitribai Phule University of Pune for the academic year 2024-2025

Ganesh Ghule: (TA-37)

Date:

Guide Name	H. O. D.
Prof. A. P. Bhalke	Dr. M. P. Borawake

ACKNOWLEDGEMENT

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ABSTRACT

This project focuses on the development of a private cloud infrastructure for Software as a Service (SaaS) within a laboratory environment using the existing Local Area Network (LAN). The system is designed with a custom-built cloud controller implemented using open-source technologies, integrated with Hadoop Distributed File System (HDFS) for scalable and fault-tolerant storage. The controller performs essential cloud operations such as segmenting files into blocks, encrypting them for security, and handling file upload and download processes to and from the cloud. This solution ensures data confidentiality, efficient storage, and seamless accessibility, serving as a foundational step toward real-time cloud-based applications in educational or small-scale enterprise environments.

Practical Title:

Setup your own cloud for Software as a Service (SaaS) over the existing LAN in your laboratory. In this assignment you have to write your own code for cloud controller using open-source technologies to implement with HDFS. Implement the basic operations may be like to divide the file in segments/blocks and upload/ download file on/from cloud in encrypted form.

Objectives:

- To set your own cloud for SaaS over existing LAN
- To implement the basic operations may be like to divide the file in segments/blocks

Hardware Requirements:

Pentium IV with latest configuration.

Software Requirements:

Ubuntu 20.04, VMwareESXi cloud

Theory:

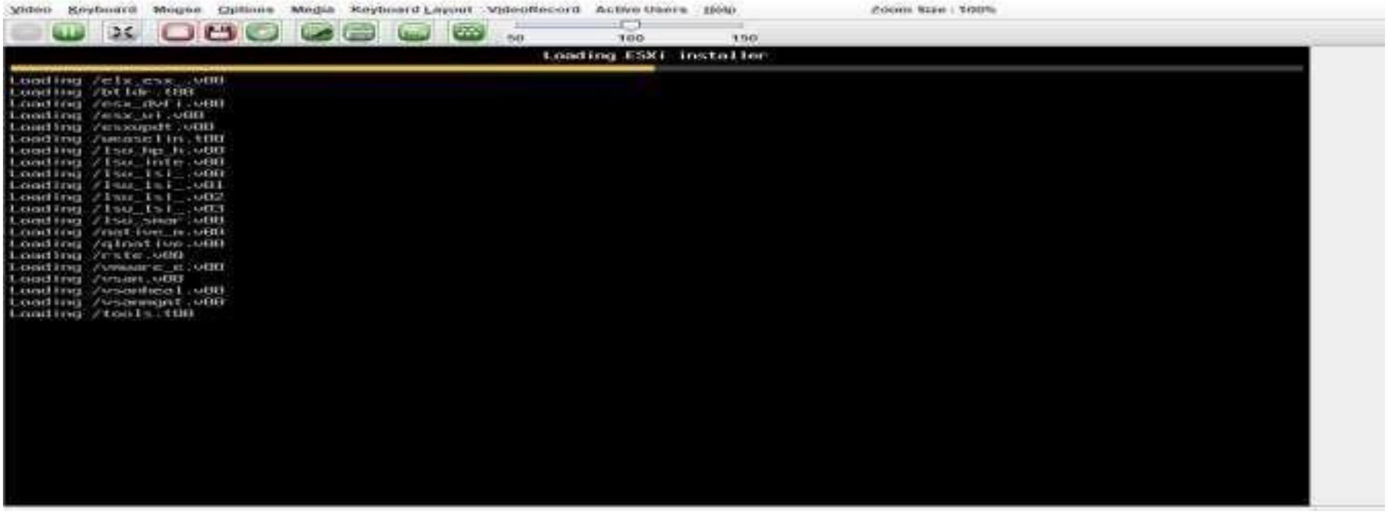
Here we are installing VMwareESXi cloud

- Host/NodeESXi installation:-
- **ESXiHardwareRequirements:-**

- ESXi6.7requiresahostmachinewithatleasttwoCPUcores.
- ESXi6.7supports64-bitx86processors
- ESXi6.7requirestheNX/XDbit to be enabled for the CPU in the BIOS.
- ESXi6.7requiresaminimumof4GBofphysicalRAM.Itisrecommended to provide atleast 8 GB of RAM to run virtual machines in typical productionenvironments.
- Tosupport64-bitvirtualmachines,support for hardware virtualization (IntelVT-xor AMDRVI) mustbeenabledonx64CPUs.
- One or more Gigabit or faster Ethernet controllers. For a list of supportednetwork adapter models.
- SCSI disk oralocal,non-network,RAIDLUN with unpartitioned space

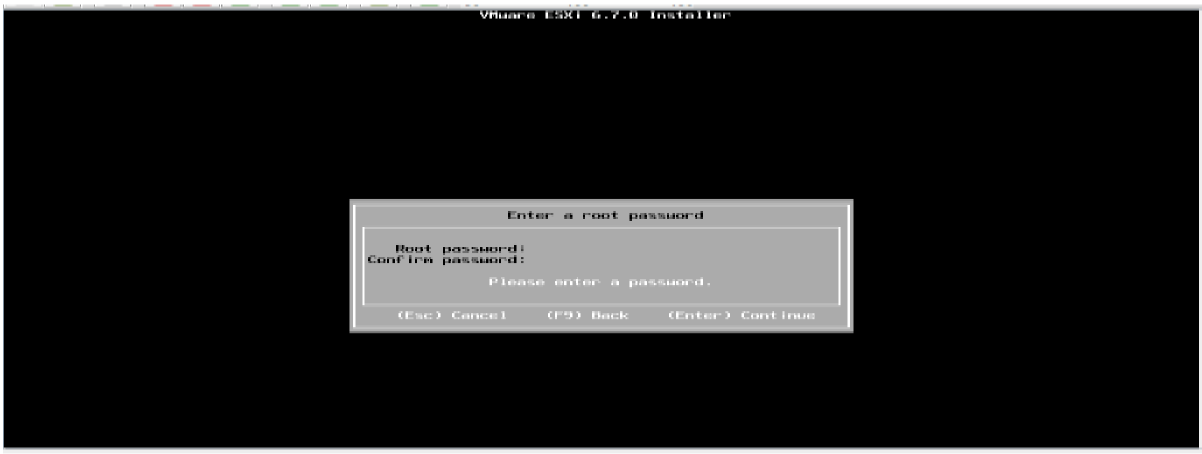
for the virtualmachines.

ForSerialATA(SATA), a disk connected through supported SAS controller or supported on board SATA controllers. SATA disks are considered remote not local. These disks are not used as a scratch partition by default be cause they are seen as remote.

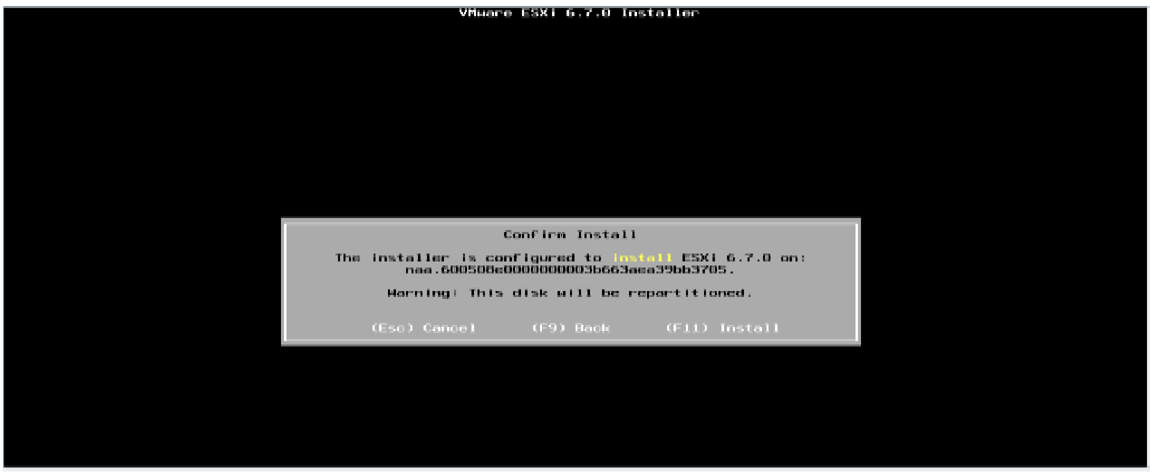


Select storage:

Select Keyboard Layout :

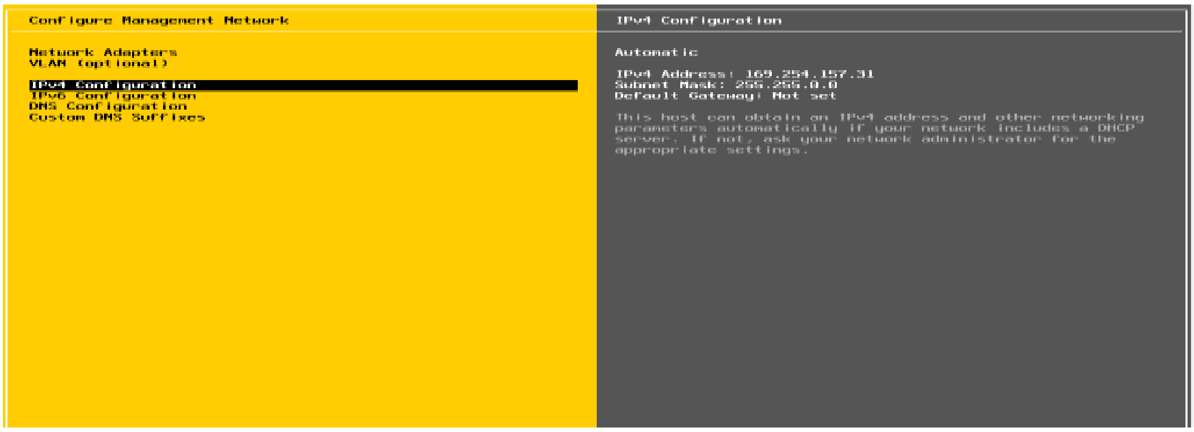


Set NodeESXi Root Password :



Installation complete (Reboot)CLII interface to configuration

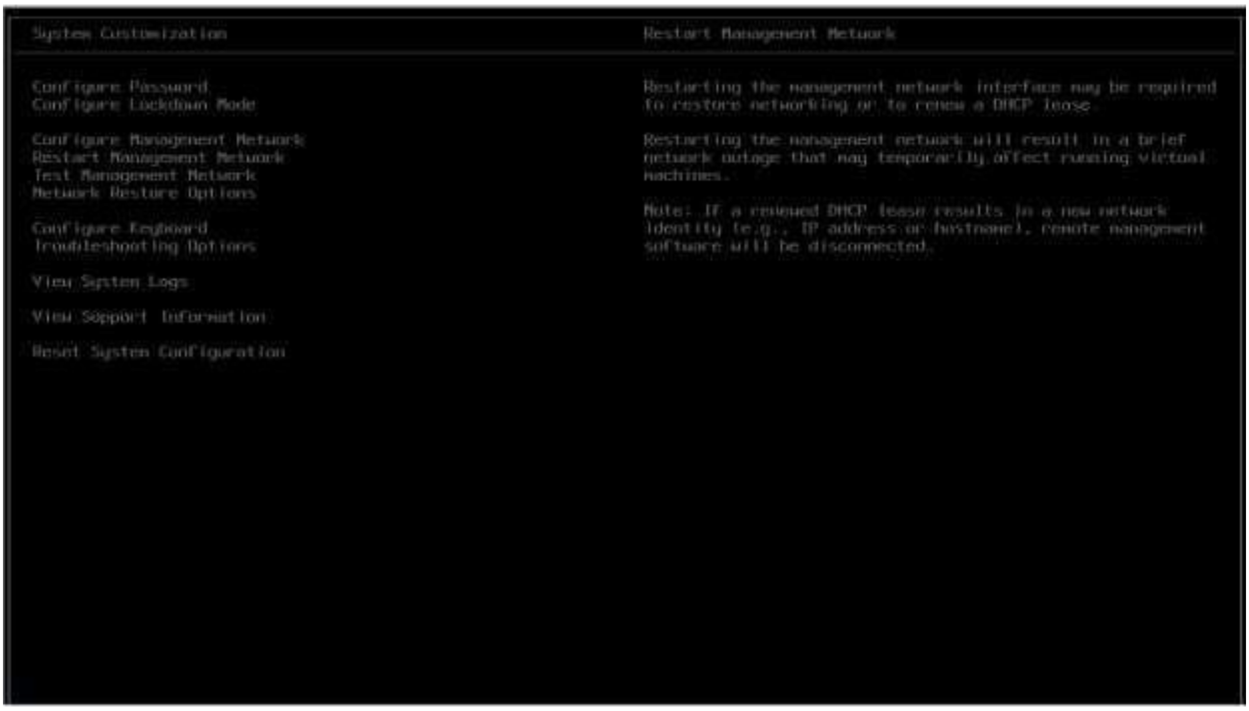
CLI Interface to Configuration:



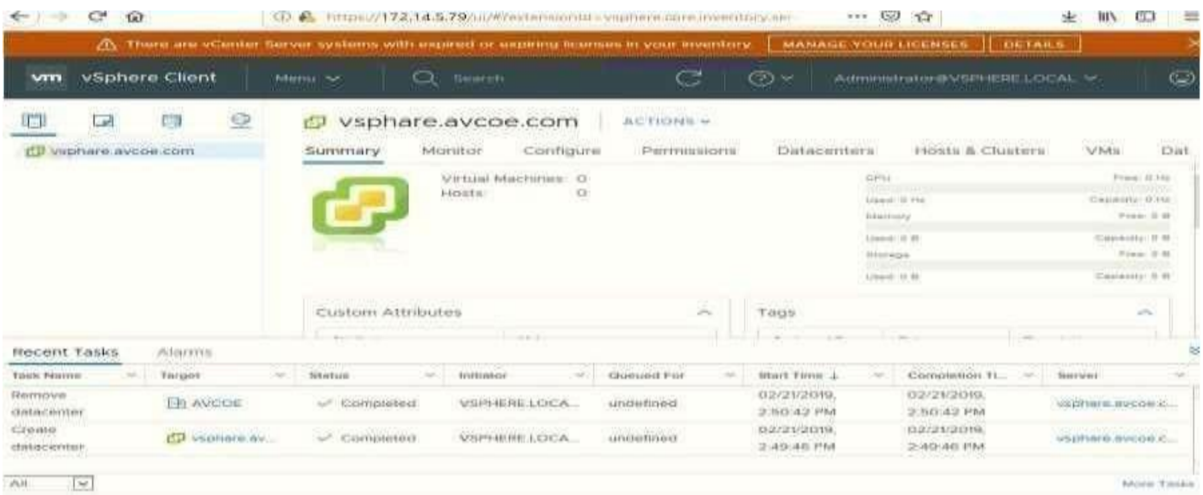
Configure Management Network

Set DNSeriver :

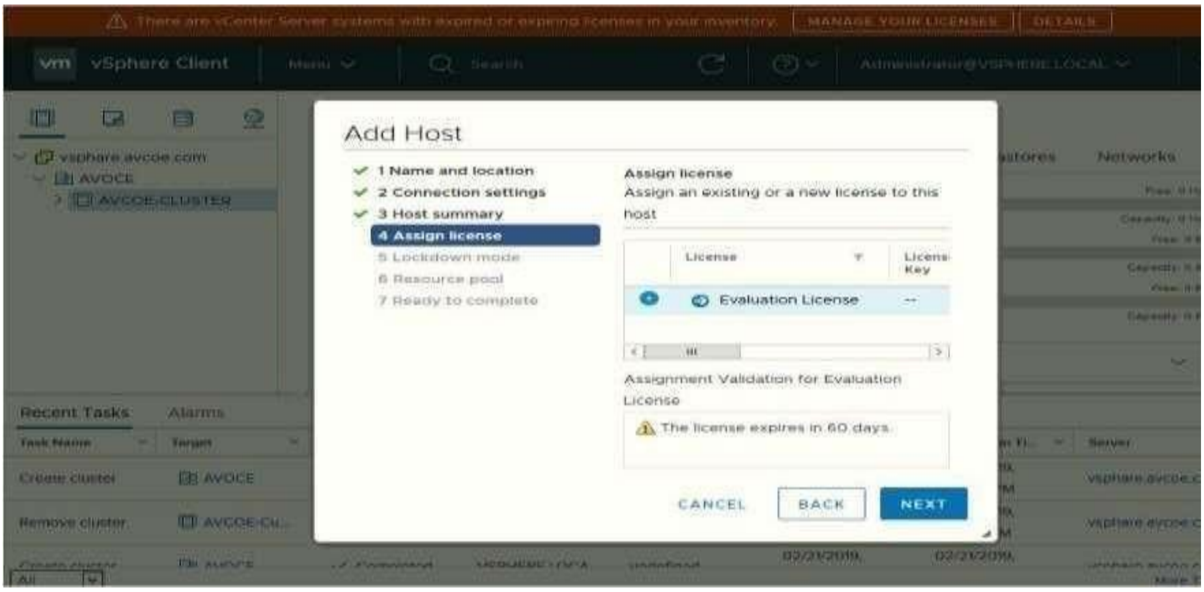
Restart Management Network



GUIAccess :

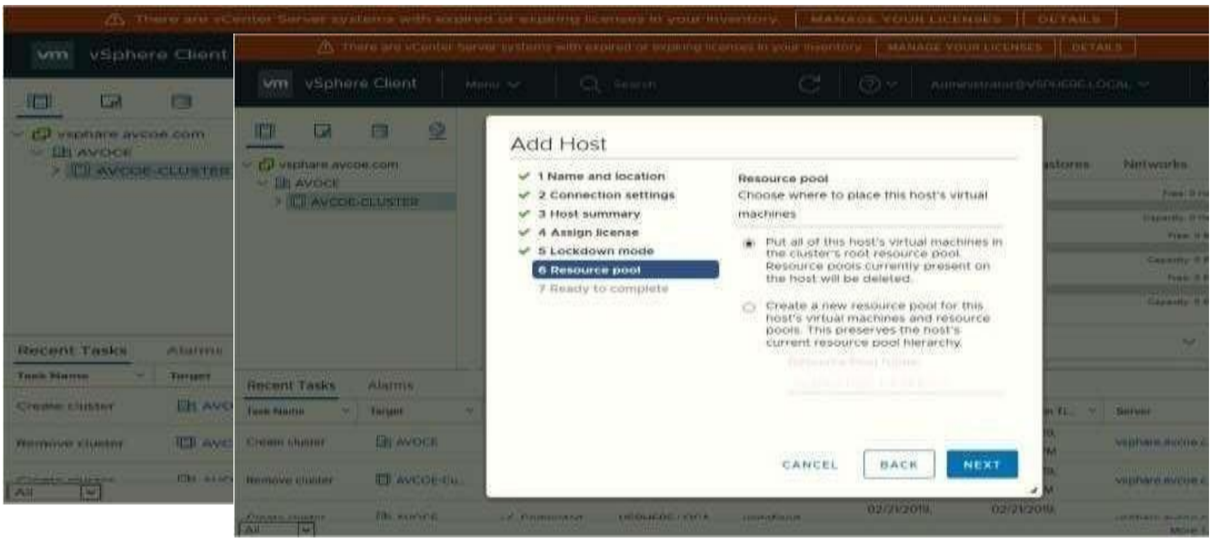


VCenter Access



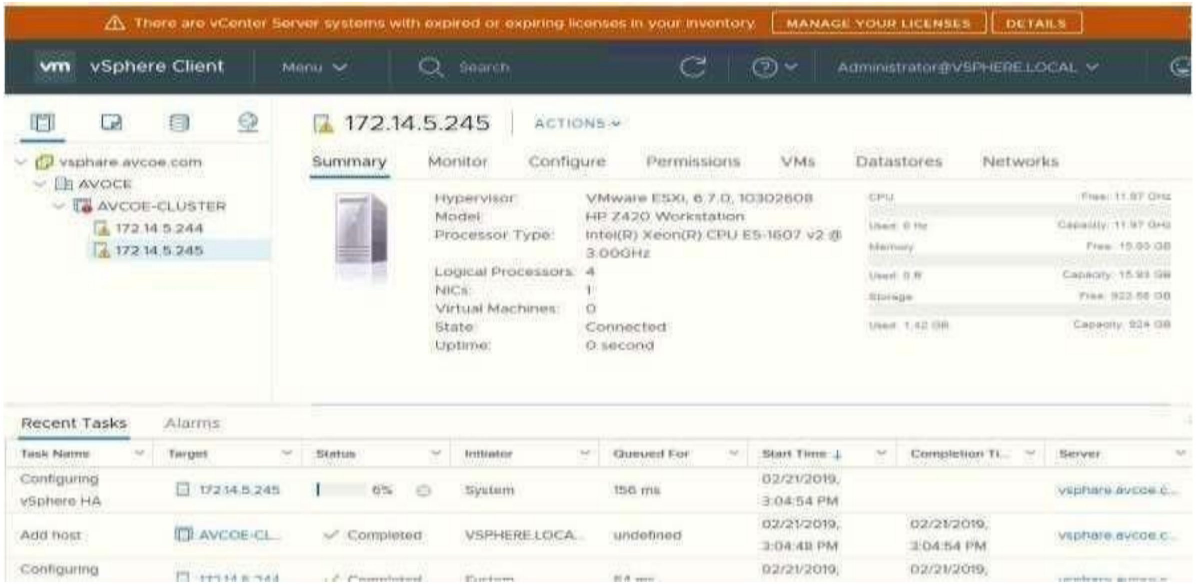
Lock Down mode:

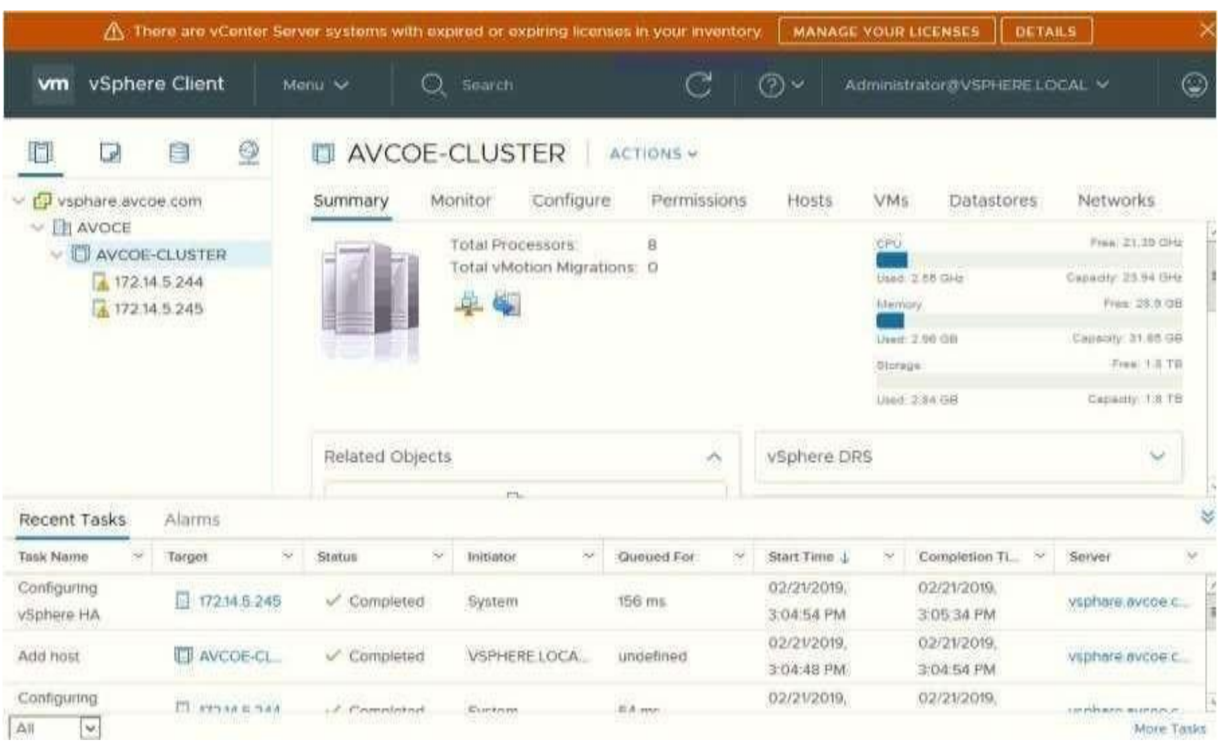
Add Host In Pool:



Host View and View Config:

Cluster View and Configuration:





Conclusion:

In this project, we successfully implemented a private cloud infrastructure within our laboratory environment using open-source technologies and integrated it with HDFS for reliable and distributed storage. By writing custom code for a cloud controller, we were able to simulate the basic functionalities of a Software as a Service (SaaS) model, including segmenting large files into blocks, encrypting them for secure transfer, and enabling upload and download operations across the local cloud setup.

This practical exercise gave us hands-on experience with cloud computing fundamentals, data security practices, and distributed file systems. It helped us understand the role of cloud controllers and storage nodes in managing file transfers, encryption mechanisms, and system performance over a LAN setup.

Additionally, we explored the configuration and management of a **vSphere Private Cloud**, which allowed us to understand how enterprise-level virtualization platforms operate. Through this, we learned about resource allocation, VM provisioning, and cloud service delivery in a controlled private network environment.

Overall, this assignment strengthened our practical knowledge in cloud computing and provided a solid foundation to explore more advanced cloud services and architectures in the future. It also enhanced our understanding of how cloud-based SaaS models can be implemented securely and efficiently in both educational and enterprise settings.